Automated Water Quality & Contamination Detection Using Computer Vision, Machine Learning, Deep Learning and Generative AI

INDIVIDUAL THESIS

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Individual component: Contamination Detection (Object Detection)

Group ID: R26-IT-005

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026 — Research design draft

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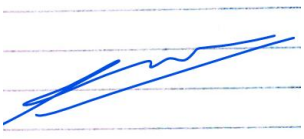
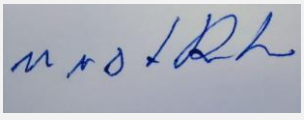
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ABSTRACT

Reliable identification of visible contamination in water samples requires methods that provide consistent classification and spatial localization under varying image-acquisition conditions. This study proposes a deep-learning-based Contamination Detection component within an integrated water-quality assessment system. The component aims to identify visual indicators of turbidity-related cloudiness, discoloration, suspected oil sheen, algae-like regions, floating debris and sufficiently resolved suspected plastic particles in water-sample photographs.

The proposed methodology involves controlled image acquisition, a documented annotation protocol, transfer learning and comparative evaluation of a YOLO-family detector and Faster R-CNN. Images will be partitioned by physical samples into independent training, validation and test sets to minimize data leakage. Discrete objects and diffuse visual conditions will be analyzed separately to account for differences in their spatial characteristics. The component will produce bounding boxes, predicted contamination categories, confidence scores and visual-severity grades defined independently of model confidence. Evaluation will examine class-wise precision, recall, mean average precision, false-positive rates, agreement with reviewer-assigned severity grades and inference latency. Robustness will also be assessed across variations in lighting, acquisition devices and container backgrounds.

The anticipated contribution is a reproducible visual-screening pipeline that integrates traceable detection outputs with the downstream Water Safety Report and Purification Advisor. As implementation and experimental evaluation remain pending, no performance claims are made. The proposed component supports preliminary screening and further investigation; photographic observations alone cannot confirm microplastic identity, quantify physicochemical contamination or establish drinking-water safety.

Keywords: water contamination, computer vision, deep learning, object detection, YOLO, Faster R-CNN, visual severity.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Mr. Buddika Harshanath, and my co-supervisor, Prof. Samantha Rajapaksha, for their guidance, constructive feedback and encouragement throughout this research.

I also thank the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for the academic support and opportunity to undertake this study.

My appreciation extends to my research group members for their collaboration, shared ideas and support in developing the overall water-quality assessment project. Their contributions have helped shape my individual component, the Water Safety Report and Purification Advisor.

Finally, I am grateful to my family and friends for their patience, understanding and continued encouragement throughout my academic journey.

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LIST OF ABBREVIATIONS

The abbreviations used in this thesis are listed below.

Abbreviation	Meaning
AP	Average precision
API	Application programming interface
CNN	Convolutional neural network
GPU	Graphics processing unit
IoU	Intersection over union
JSON	JavaScript Object Notation
mAP	Mean average precision
NMS	Non-maximum suppression
R-CNN	Region-based convolutional neural network
ROI	Region of interest
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Overall project context

The overall research project proposes an integrated water-quality assessment system comprising four complementary components: photographic contamination detection, physicochemical water-quality classification, microorganism detection through microscopy, and automated report generation with purification guidance. By combining these distinct sources of evidence, the system aims to support a more comprehensive interpretation of water-quality conditions.

This individual thesis focuses on the photographic contamination detection component, which applies deep-learning-based computer vision to identify and localize visible contamination indicators in water-sample images. Its outputs complement physicochemical measurements and microscopic findings and are communicated to the reporting component through a structured interface. Each image-derived finding is represented as a visual prediction, with its provenance and associated uncertainty preserved to support transparent interpretation within the integrated system.

1.2 Individual component context

The component will accept a photograph associated with a water sample, check whether the image is suitable for analysis and apply a CNN-based object detector. The primary outputs are contamination category, bounding-box coordinates, model confidence and visual severity. A YOLO-family detector [1] is proposed as the primary candidate, with Faster R-CNN [2] as the comparison model.

The requested classes include both discrete objects and diffused appearances. Floating debris can be boxed as an individual object; discoloration may occupy most of a sample. The annotation and evaluation design therefore distinguishes object instances from visible-condition regions without treating their boxes as equivalent physical measurements.

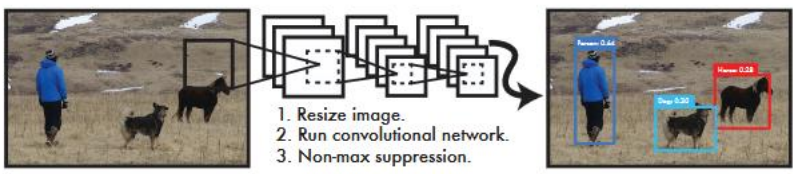


Figure 1: The YOLO Detection System

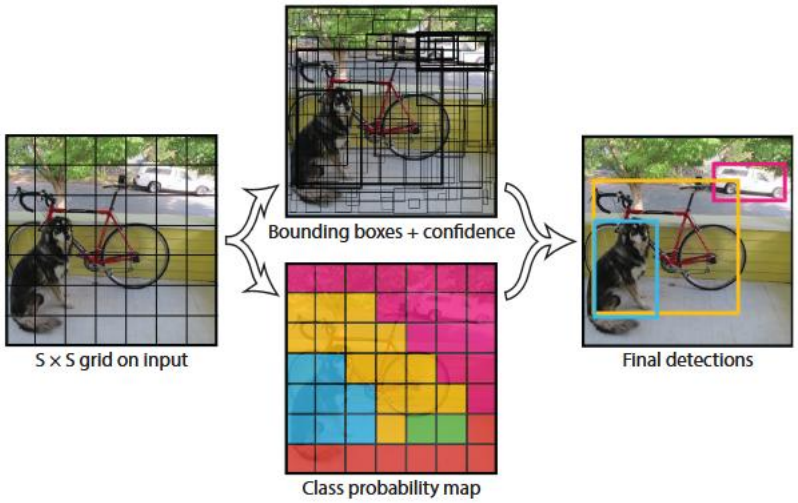


Figure 2: The Model YOLO

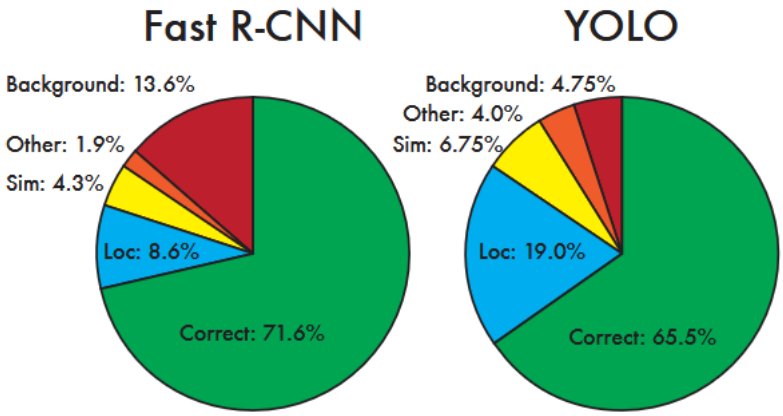


Figure 3: Error Analysis: Fast R-CNN vs. YOLO

1.3 Problem statement

The project needs a repeatable method for identifying visible contamination and showing where the supporting visual evidence appears. Manual inspection may vary across observers and capture conditions, while a simple image-level label does not indicate the relevant region. A detector must also cope with reflections, shadows, container markings and background material that resemble contamination.

The research problem is to determine whether an appropriately trained detector can localize and classify the defined visual categories under specified image-acquisition conditions, and whether its output can support a reproducible visual severity estimate. Performance must be established on independent samples rather than repeated photographs of training samples.

1.4 Research gap

General-purpose object-detection architecture provides established methods for identifying and localizing visual targets [1], [2]. However, performance demonstrated on general benchmark datasets does not establish their effectiveness for water-sample analysis. This application requires domain-specific class definitions, consistent annotation procedures and evaluation under variations in illumination, container appearance, background conditions and target size. Such factors may influence the distinction between contamination indicators and visually similar features, including reflections, bubbles and container markings.

A central methodological challenge concerns the distinction between discrete objects and diffuse visual conditions. Floating debris may be represented by individual bounding boxes, whereas cloudiness, discoloration and surface sheen may extend across substantial portions of the visible water region without clearly defined boundaries. Applying a uniform annotation and evaluation strategy to these categories may therefore obscure differences in localization performance. The proposed study addresses the need for a framework that explicitly distinguishes these categories and evaluates their detection characteristics separately.

Further research question concerns the relationship between detected visual evidence and contamination severity. Model confidence indicates the strength of a prediction but does not establish the extent of contamination or its associated health risk. Similarly, bounding-box coverage provides only an approximate representation of visible extent, particularly for diffuse conditions. The study therefore proposes a separately defined visual-severity method, evaluated against independent reviewer assessments, to investigate whether detection outputs can support consistent and interpretable severity reporting.

Accordingly, the project-specific research gap concerns the integration and evaluation of contamination localization, differentiated handling of discrete and diffuse indicators, and independently assessed visual severity within a traceable water-quality reporting workflow. The intended contribution is an application-specific methodology and empirical evaluation rather than new object-detection architecture. Broader claims of novelty will depend on a more comprehensive review of comparable water-image analysis systems.

1.5 Aim and objectives

Aim: To develop and evaluate a CNN-based object-detection component that locates and classifies visible water-contamination indicators and reports their visual severity within a defined capture setting.

Objectives:

1: Define image-acquisition procedures, contamination labels, annotation rules and reference severity grades.

2: Curate a licensed, traceable dataset and create independent training, validation and test partitions.

3: Fine-tune and compare a specified YOLO-family detector and Faster R-CNN for localization and classification.

4: Develop a visual-severity method and evaluate it independently of detector confidence.

5: Integrate predictions into the group system and assess detection quality, robustness, latency and failure handling.

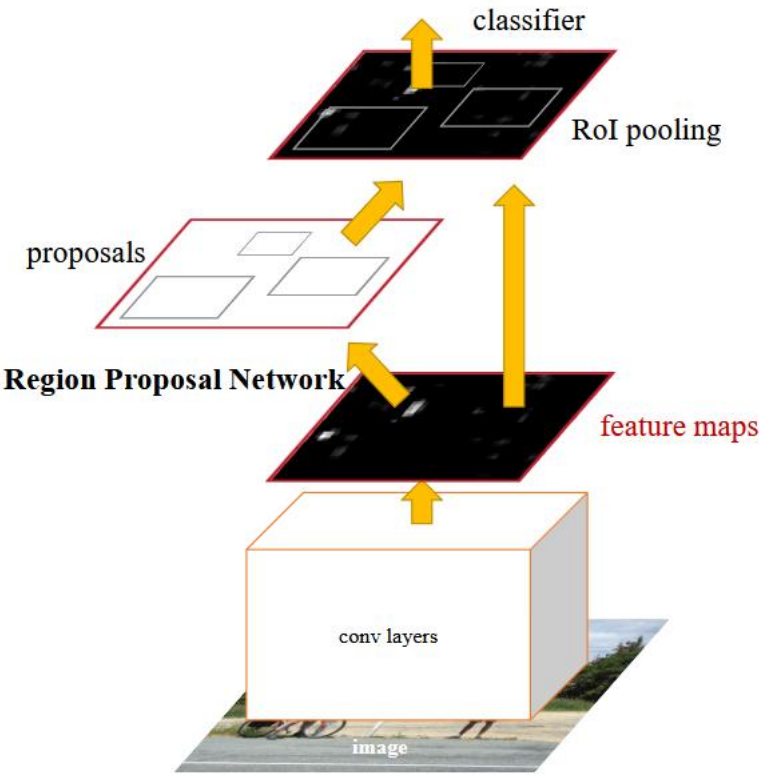


Figure 4: Faster R-CNN

1.6 Research questions or hypotheses

Q1: How accurately do the two candidate detectors locate and classify the defined contamination indicators on independent water samples?

Q2: How does performance differ between discrete objects and diffuse conditions, and between large and small visible regions?

Q3: What is the effect of lighting, container background and acquisition-device changes on false positives and missed detections?

Q4: How closely does the proposed visual-severity output agree with independent reviewer grades?

Q5: What accuracy–latency trade-off is obtained by the candidate models under the intended deployment environment?

The working expectation is that transfer learning and domain-specific training will improve detection over an unadapted detector. No direction of superiority between YOLO and Faster R-CNN is assumed before evaluation.

1.7 Scope limitations and assumptions

The initial scope controls photographs of collected water samples with a documented viewing geometry, background and visible water region. Field images of rivers or reservoirs are a separate domain and will not be pooled into the main test set without explicit evaluation. The system will not measure dissolved contaminants, determine microbial species or prescribe treatment.

Turbidity-related cloudiness and discoloration will be appearance labels, not instrument readings or chemical diagnoses. Oil sheen and algae bloom will be visually suspected categories unless reference evidence confirms the material. Ordinary photographs will not establish the chemical identity of microplastics. Only sufficiently resolved candidate particles will be included under a clearly qualified visual label; a confirmed microplastics claim requires a separately validated imaging and reference-identification workflow [3].

The prototype assumes permission to acquire or reuse images and access to annotation reviewers. Empty detections mean that the configured model found no supported visible target, not that the water is safe.

1.8 Individual contribution

The planned individual contribution includes the capture protocol, class ontology, annotated image dataset, leakage-resistant split manifest, detector training and comparison, severity estimator, inference endpoint and evaluation records. The candidate will own the image-based component while the team’s numerical, microbial and advisory modules remain separate contributions.

Repository history, annotation records and experiment logs should later substantiate actual ownership. This draft does not imply that a trained model or dataset has already been produced.

1.9 Thesis structure

Chapter 2 reviews the detection foundations and scope limitations. Chapter 3 describes data preparation and evaluation. Chapter 4 specifies the solution architecture and severity design. Chapter 5 covers planned implementation and testing. Chapter 6 identifies the pending results, Chapter 7 discusses contributions and limitations, and Chapter 8 concludes with objective status and future directions. Appendices provide annotation and test instruments plus AI-use disclosure.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This study adopts a focused narrative review to establish the conceptual and methodological foundations of image-based water-contamination detection. The initial review covers primary publications on YOLO [1] and Faster R-CNN [2], the COCO dataset paper [4], and an authoritative report on microplastics. Search terms address object-detection architectures, annotated image datasets and microplastics identification. The selected literature is examined for its relevance to contamination localization, annotation design and performance evaluation.

The review is not presented as a systematic literature review. Its scope will be extended to include primary studies on water-surface contamination, image-based turbidity assessment, algae, oil sheen and small-particle detection. Studies will be assessed according to dataset characteristics, image-acquisition conditions, reference annotations, evaluation metrics and reported limitations. Quantitative comparisons will be restricted to studies with sufficiently comparable tasks and evaluation protocols. Research conducted under substantially different conditions will inform the contextual discussion without treating its reported performance as directly comparable.

2.2 Domain and theoretical foundations

An object detector estimates both the class and spatial extent of a visual target. A bounding box is an image rectangle rather than a physical boundary or contaminant concentration. A convolutional neural network (CNN) learns image features during training; transfer learning adapts pretrained parameters to the target dataset [5].

Intersection over union (IoU) measures overlap between a predicted and reference box: intersection area divided by union area [1]. Detection confidence and severity describe different quantities. Confidence concerns a model prediction; severity in this thesis describes a defined visual extent or appearance grade. Neither is a probability of drinking-water safety.

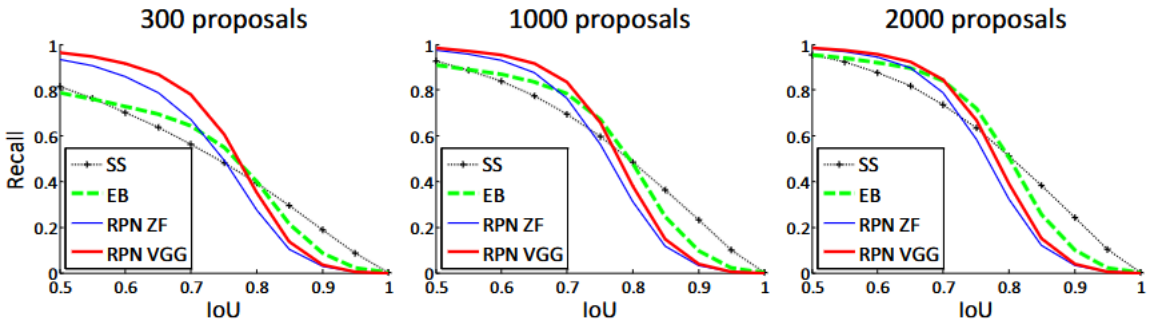


Figure 5: Recall vs. IoU overlap ratio on the PASCAL VOC 2007 test set

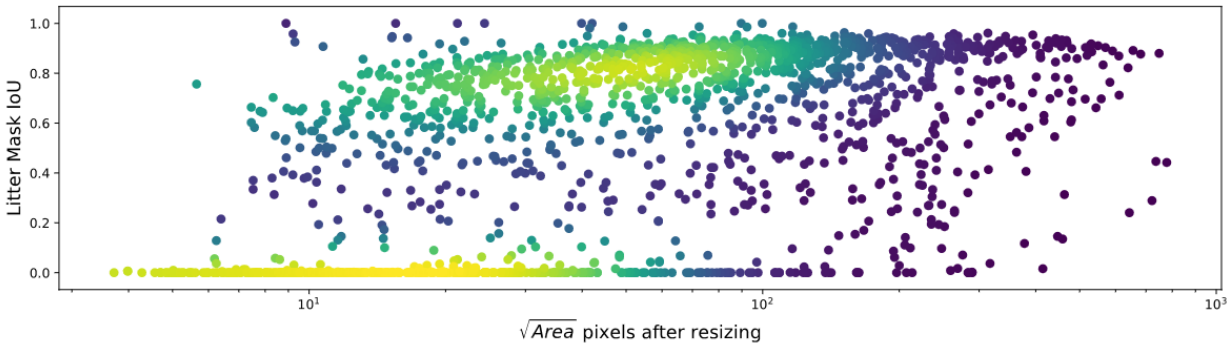


Figure 6: Annotation bounding box area vs best matched IoU. Color represents density of scattered points

2.3 Related research

Redmon et al. [1] introduced YOLO as a unified detector that predicts boxes and class probabilities from an image in one network evaluation. That work motivates a single-stage baseline, but its reported speed and accuracy belong to its original datasets and hardware. Any contemporary YOLO implementation used here must be identified by exact architecture, checkpoint and software revision.

Ren et al. [2] introduced Faster R-CNN with a region proposal network sharing convolutional features with the detection network. It offers a contrasting proposal-based architecture for the experiment. This thesis will compare implementations under common data and evaluation conditions rather than assuming one is superior.

Lin et al. [4] describes COCO as an object-recognition dataset with instance annotations and benchmark tasks. Such datasets can supply transferable visual representations, but do not provide the project's water-contamination ground truth. WHO's microplastics report [3] provides context for the need to distinguish visible particles from verified material identity.

Lin et al. [6] proposed feature pyramid networks to construct multi-scale feature representations for object detection. This work informs the discussion of target-size variation; its benchmark results do not establish small-particle detection performance in water samples.

Lin et al. [7] introduced focal loss to address foreground–background imbalance in dense object detection by reducing the contribution of well-classified examples. This provides methodological context for training imbalance, without implying that focal loss has been selected or evaluated in this project.

Proença and Simões [8] introduced TACO, an annotated dataset for litter detection and segmentation. Its annotation practices provide a relevant reference for visible debris, although its images do not constitute water-sample contamination ground truth.

Biermann et al. [9] investigated the identification of floating plastic patches using optical satellite imagery. The study provides contextual evidence for image-based aquatic debris analysis, but its imaging scale and spectral information differ from the controlled water-sample photographs proposed here; its results are therefore not a directly comparable benchmark.

2.4 Existing systems products and techniques

Candidate approaches include manual image inspection, classical colour or texture rules, image-level classification, object detection and pixel-level segmentation. Inspection provides useful reference annotations but needs documented reviewer consistency. Colour rules can serve as a limited diffuse-condition baseline but must be tested against lighting and background changes. Image classification identifies an overall condition without localizing discrete objects.

Object detection matches the requested bounding-box output. Segmentation is a possible extension when the actual contaminated area is required, because a box encloses background pixels as well as the target. The core study will keep object detection central and label any box-derived coverage value as an approximation.

2.5 Gap synthesis and conceptual framework

The proposed conceptual framework addresses the project-specific need to connect contamination localization with interpretable visual-severity assessment and reporting. It links image-acquisition quality and annotation consistency to detection performance, followed using localized visual evidence to estimate severity. Capture conditions, model architecture and preprocessing strategies constitute the principal experimental factors. Outcomes will include class-wise detection performance, false-positive rates, agreement with independently assigned severity grades and inference latency. This framework supports an examination of how data quality and methodological choices influence the reliability of the component's outputs.

The analysis will distinguish discrete object instances, such as floating debris, from diffuse visual conditions, such as cloudiness and discoloration. These categories present different localization challenges and require distinct interpretation of bounding-box overlap and spatial extent. A prediction covering the entire water region, for example, does not demonstrate the same localization capability as the accurate detection of a small particle. Performance will therefore be reported separately for both groups alongside aggregate results, ensuring that category-specific strengths and limitations remain explicit. Visual severity will also be evaluated independently of detection confidence to preserve the distinction between prediction certainty and the extent of the observed condition.

3. RESEARCH METHODOLOGY

3.1 Research design

This study proposes a design-and-evaluate research methodology to develop and assess an image-analysis service for visible water-contamination detection. The research artefact will comprise an annotated image dataset, a trained object detector and a visual-severity assessment module. Development will begin with a reviewed pilot dataset to refine image-acquisition procedures, annotation guidelines and severity criteria. Following this preliminary stage, a fixed experimental protocol will guide controlled model comparisons and evaluation using a held-out test set.

The research design distinguishes empirical model evaluation from software verification. Detection experiments will examine the models' ability to generalize independent water samples, while severity evaluation will assess agreement with independently assigned reference grades. Unit and contract tests will verify the preservation of sample and image identifiers; the correctness of bounding-box coordinate transformations and compliance with the defined output schema. These complementary activities will assess predictive performance and implementation correctness separately. Data collection, model training and experimental evaluation remain planned activities at the current stage of the research.

3.2 Data participants or study materials

Materials will include original water sample photographs and suitable externally licensed images with documented provenance. An initial planning target is approximately 1,500–3,000 images, subject to feasibility and reviewer capacity. This is a recruitment target, not an existing dataset or a statistical sufficiency claim. Report independent physical sample counts as well as image counts; many views of one sample do not provide many independent samples.

Acquire images across controlled variations in illumination, device, container and contamination appearance. Include negative examples and hard negatives such as glare, bubbles, shadows and container marks. Archive sample ID, capture session, device, resolution, viewing distance, background and any verified preparation or reference information. Avoid deliberately introducing hazardous materials for image collection without approved facilities and procedures.

Table 3.1 defines the proposed labels. Two reviewers should independently annotate a pilot subset; disagreements will be resolved before expanding the dataset. Ambiguous material identity will be preserved as uncertainty rather than forced into a confident class label.

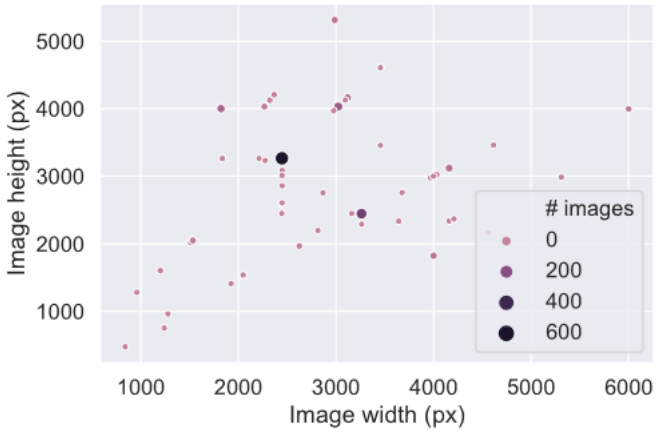


Figure 7: Distribution of image resolutions

Table 1: Class and annotation policy

Requested category	Proposed annotation	Interpretation
Turbidity	Turbidity-related cloudiness region or water ROI	Visual condition; not a numerical turbidity reading.
Discoloration	Affected water region; full ROI if uniform	Appearance label, with capture conditions controlled.
Oil sheen	Visible sheen region	Suspected oil-like appearance unless verified.
Algae bloom	Visible algae-like patch or region	Suspected appearance; no species or toxin claim.
Floating debris	One tight box per separable visible item	Object instance; group rules documented for dense clusters.
Microplastics	Resolvable suspected plastic particle only	Exploratory label; ordinary photos do not confirm identity.

3.3 Experimental or study procedure

First, pilot the capture protocol and annotation guide. Define a water region of interest (ROI) so background surfaces are not graded. Review blur, exposure and resolution before annotation. Label each eligible object or condition region and preserve independent visual-severity grades. Record uncertain cases for adjudication.

Split the dataset approximately 70%/15%/15% into training, validation and testing at the physical-sample or capture-session group level before augmentation. Keep duplicate and near-duplicate images together and avoid source-specific leakage. Where enough data exists, reserve an additional device or site for external-domain testing. Reports realized class counts in every split.

Fine-tune one specified YOLO-family detector and one Faster R-CNN implementation from documented pretrained checkpoints [5]. Use the same split, label policy and evaluation code. Allocate comparable tuning budgets, log architecture-specific hyperparameters and select checkpoints using validation outcomes. Run at least three training seeds when feasible. Optimizer, learning rate, batch size, input size and stopping rule will be chosen on development data and frozen before testing.

Compare a standard-preprocessing baseline with a carefully limited augmentation variant. Avoid strong color transformations that change the meaning of discoloration or sheen labels. Evaluate small-target preservation at two feasible input resolutions. Select confidence and suppression settings on validation data only. Test once the configuration is fixed, retaining full predictions for independent scoring.

3.4 Measures and evaluation metrics

Table 3.2 specifies the planned metrics. Match predictions to references within the same class using one-to-one assignment at the stated IoU threshold. Report per-class results and the discrete/diffuse groups separately. Background and unlabeled regions must follow the agreed annotation policy to avoid treating missing annotations as false detections.

Average precision summarizes the precision–recall curve. The study will report AP at IoU 0.50, and mAP averaged across IoU thresholds 0.50 to 0.95 in steps of 0.05, with the evaluator settings recorded. Severity will be scored against independent ordinal grades, including an unavailable output where evidence is inadequate [4].

Table 2: Evaluation measures

Metric	Reporting requirement
Precision / recall / F1	At validation-selected score threshold and declared IoU; per class.
AP50 and mAP50–95	Class-wise and macro averages; discrete/diffuse results separated.
False positives per image	Include accepted negative and challenging negative photographs.
Size and domain robustness	Stratify by pixel size, device, lighting and collection session.
Severity agreement	Confusion matrix, macro-F1, ordinal error, weighted agreement and coverage.
Operational performance	Median/p95 latency, hardware, resolution, batch size and memory usage.

3.5 Data analysis

Report means and variability across training seeds, and bootstrap uncertainty by physical sample rather than by individual image. Use paired test images for comparing model configurations. Stratify error analysis by class, region size, capture condition and image quality. Do not select the best test-set run and present it as the typical result.

For severity, report the confusion matrix, macro-F1, ordinal mean absolute error and weighted agreement with reviewers. Publish reviewer agreement before adjudication. Include both conditional severity accuracy for correctly detected cases and end-to-end severity coverage so missed detections do not disappear from the analysis. Investigate false positives on uncontaminated-looking images and missed small targets with representative examples.

3.6 Ethics privacy security and responsible AI

Use images acquired with permission and confirmed external dataset licences. Remove irrelevant identifiers, faces and location metadata from shared artefacts. Any human annotation or usability study will follow institutional consent and review requirements. Collecting real samples must follow the project's approved handling procedures; the model does not justify unsafe collection activities.

Responsible presentation requires qualified class labels and visible limitations. Do not convert no detection into a safe-water label, predicted algae into species confirmation, or a plastic-like particle into chemically verified microplastic. Preserve image provenance and record unsupported capture conditions. Prevent uploaded images from executing code and validate file content and dimensions at the service boundary.

3.7 Reproducibility and research artefacts

Retain the image manifest, licences, sample-group IDs, split files, annotation version, class maps, preprocessing parameters, code commit, environment lock file, pretrained checkpoint hash, training seeds and final weights. Evaluation artefacts will include predicted boxes, scores, severity values, metrics, inference timings and failure examples.

When images cannot be redistributed, provide permitted metadata and reconstruction instructions. The final thesis should identify the exact evaluator and any deviations from standard settings. Appendix A provides the proposed annotation record and test cases; Appendix B records document-drafting assistance.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements are image upload with sample identity, quality screening, water ROI identification, contamination detection, original-image coordinate recovery, class/confidence output, visual-severity estimation, annotated preview and structured delivery to the group reporting service. Users must be able to distinguish a rejected image from an accepted image with no detections.

Non-functional requirements include repeatable preprocessing, traceability, bounded input size, graceful failures and measured performance on the target hardware. Accuracy and response-time acceptance targets will be agreed after pilot evidence. No unmeasured real-time capability or minimum accuracy is asserted.

4.2 Component architecture within the complete solution

The component has an offline training path and an online inference path. Offline processing comprises collection, review, annotation, group-wise splitting, training, validation and model registration. Online processing comprises file validation, image-quality assessment, ROI selection, model-specific resizing, inference, postprocessing, coordinate restoration, severity scoring and output packaging.

The system dashboard sends an image and sample ID to the detection service. The service returns the fields in Table 4.1 to the Water Safety Report and Purification Advisor. Numerical water-quality and microscope modules may analyse the same sample separately; this detector does not alter their results. The downstream report must retain the distinction between visible predictions and confirmed measurements.

Table 3: Output contract

Field	Meaning
sample_id / image_id	Traceable input identity; unchanged across modules.
schema_version / model_version	Contract and exact trained-model revision.
image_width / image_height	Original image dimensions for all returned boxes.
status / quality_warnings	Distinguish valid empty results from rejected or failed analysis.
detections[]	Class ID, qualified label, confidence and original-pixel box.
severity / severity_method	Class-specific visual grade or unavailable; method and version.
evidence_scope	Visual prediction, not laboratory confirmation or potability.
latency_ms / audit_id	Measured processing time and trace reference.

4.3 Design alternatives and decisions

A YOLO-family candidate is proposed for the primary implementation, while Faster R-CNN provides an architecture comparison. Final selection depends on measured per-class accuracy, failure patterns, latency and deployment constraints. The name “YOLO” alone is not a reproducible model specification; record variant, backbone, input resolution, checkpoint and licence.

Bounding boxes are retained to match the component requirement. Where the visible phenomenon is diffuse, the label policy will specify the affected region or full water ROI. Image-level presence scores will accompany analysis of these conditions. If adequate region boundaries cannot be agreed by reviewers, report that limitation and treat segmentation or a separate appearance classifier as future work rather than pretending localization is precise.

4.4 Detailed design

Each detection will contain class_id, display_label, box_xyxy, confidence and evidence_scope. Pixel boxes will use the original image dimensions; JSON output will document whether maximum coordinates are exclusive. Preprocessing must retain scale and padding values so boxes can be mapped back correctly. Invalid or out-of-bound boxes will be rejected or clipped under a documented rule.

For discrete targets, proposed severity is a class-specific visual-extent grade. Let W be the accepted water ROI and B be the union of retained same-class boxes clipped to W. The proxy r = area(B)/area(W) avoids double-counting overlapping boxes. Because rectangles include background, r is box coverage, not true contaminant area, volume or concentration. Reference reviewers will assign low, moderate or high visual extent using documented rubric; grade thresholds will be calibrated on development data only.

For diffuse cloudiness and discoloration, full-ROI boxes cannot estimate severity because their area can remain constant across appearance levels. Instead, a separate ordinal severity estimator is proposed using the cropped water ROI and reviewer grades, with capture conditions standardized. It may share image features with the detector or use a small independent classifier; this is a design addition requiring its own evaluation. Without adequate labels, return severity unavailable.

The model score must never be reused as severity. A low-confidence detection may have large visual extent, while a high-confidence detection may be small. Quality failures, unknown class appearances and unresolved grading will generate explicit warnings. Status values will distinguish detections_present, no_detection, rejected_image and inference_failed.

4.5 Standards and professional practice

Maintain an annotation manual, model card, dataset documentation and change log. These will record intended use, capture conditions, class definitions, known failure modes and limitations. Cite the source and licence for reused images, pretrained weights and software. General benchmark success does not establish suitability for this application.

The interface should use readable labels and show confidence separately from visual severity. Changes to labels, coordination conventions or model outputs must update the schema version and trigger regression tests. No formal laboratory, regulatory or drinking-water certification is claimed.

5. IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The proposed development environment uses Python, a deep-learning framework such as PyTorch, an image-processing library, an annotation tool and a versioned inference API. Candidate detector implementations must be chosen after confirming architecture availability, licence and compatibility. Training will use a GPU where available; inference will be measured on the actual intended deployment hardware.

Record exact dependency versions, driver/runtime information and model configuration when implementation begins. This section specifies a candidate environment and does not claim those packages or models have been installed, trained or benchmarked.

5.2 Component implementation

The planned implementation sequence is to create the class map and image manifest, implement image and annotation validation, build the sample-wise splitter, configure training, develop inference and coordinate restoration, add severity estimation, and expose the output endpoint. Each stage will produce a small reviewable artefact before the full experiment.

Training logs will retain validation curves and checkpoint-selection decisions. Inference will return machine-readable predictions and an optional overlay image. Detection labels, thresholds and severity rules will reside in versioned configuration rather than being scattered through dashboard code. The supplied proposal contains no source code, so all steps remain planned.

5.3 Integration with the other components

The dashboard will submit one sample ID, one image ID, acquisition metadata and the image. The response will include the model and schema versions, original dimensions, quality status, detections, severity method and processing time. Repeated processing should retain the original record and identify the new model version instead of silently replacing provenance.

The downstream advisor must not interpret detector confidence as overall water-safety confidence. It should state when suspected contaminants need confirmation and when image quality limits interpretation. Integration tests will use mocks before actual group modules are available, with sample IDs checked throughout the pipeline.

5.4 Verification and software testing

Software tests will validate corrupt-file handling, size limits, ROI bounds, box conversions, resizing and padding reversal, coordinate clipping, duplicate suppression and JSON schema conformance. Synthetic geometrical fixtures will test coordinate arithmetic independently of the trained model. Model tests will use held-out labelled photographs and challenging negative examples.

Severity tests will confirm that overlapping boxes are not double-counted and that full-ROI diffuse boxes do not automatically become a high grade. End-to-end tests will distinguish no detection from rejected images and inference failures. Appendix A gives representative expected behaviours; no passing test results are claimed in this draft.

5.5 Deployment and operation

Begin with a restricted research deployment and archive every released checkpoint with its configuration. Store uploads securely, apply retention limits and separate any training-data consent from routine inference. Apply file validation and resource limits before decoding large images.

Monitoring will record rejection frequency, latency, model version and user-reported false detections. New cameras or substantially different sample setups will trigger targeted evaluation. Model upgrades will require regression comparison and rollback capability. Operational performance targets will be set using measured data rather than inherited benchmark claims.

6. RESULTS AND VALIDATION

6.1 Research results

No trained-model evaluation was provided. Therefore AP, mAP, precision, recall, severity agreement and statistical comparisons remain unreported. Table 6.1 specifies the evidence needed for the final results chapter.

After testing, report dataset and independent-sample counts, class balance, acquisition conditions, reviewer agreement and model settings before presenting the scores. Include all planned comparisons and report poorly performing classes. Any excluded or unlearnable class must be explained rather than removed silently from aggregate metrics.

Table 4: Results reporting matrix

Evidence	Expected final record	Current status
Dataset	Independent samples, images, labels and split counts	Not collected/provided
Detection	Per-class scores for both candidates and seeds	Not evaluated
Severity	Reference agreement and end-to-end coverage	Not evaluated
Robustness	Device, lighting, background and size strata	Not evaluated
Software / integration	Test logs, example outputs and report trace	Not executed
Performance	Measured latency, memory and hardware configuration	Not measured

6.2 Product and technical results

Technical results remain pending. Required evidence includes a running inference service, verified coordinate recovery, sample overlays, numerical severity tests, latency distributions, memory usage and model-file size. Measure model-only inference time separately from end-to-end upload, decoding, preprocessing and output generation.

A visually convincing overlay is an illustration, not a performance result. Product readiness requires documented failure handling and compatibility with the team's sample and result schemas.

6.3 User or stakeholder validation

No stakeholder study or reviewer outcome is available. A proposed formative study will ask domain reviewers to assess label meaning, region boundaries and visual severity. Optional target-user tasks

will assess whether users understand predicted findings and recognise that no visible detection does not establish water safety.

Record actual recruitment, consent, participant characteristics and procedure if the study proceeds. Do not invent reviewer approval, user quotations or satisfaction statistics. Small convenience samples should be described as formative rather than representative.

6.4 Integration results

End-to-end integration has not been demonstrated. The final thesis should show a trace from an uploaded sample image through detection and severity outputs to the advisor report. Verify that original image coordinates align with the displayed overlay, sample IDs remain unchanged and suspected labels retain their qualifications.

Record success and failure cases, including rejected images, timeouts and missing acquisition metadata. Confirm that downstream modules receive explicit status fields rather than interpreting an empty result as a complete safety assessment.

6.5 Summary of findings

At this stage, the supported outcome is the specification of a detection component, annotation policy, a severity design and an evaluation protocol. No improvement in detection accuracy, inspection speed or user understanding has been demonstrated. Those conclusions require the measurements identified in this chapter.

7. DISCUSSION

7.1 Interpretation of findings

Interpretation must follow the planned evaluation. Strong overall mAP with weak small-particle recall would support only part of the intended scope. High diffuse-region box overlap may reflect the annotation policy rather than precise localization. Better severity scores conditional on correct detections could conceal many missed cases, so end-to-end coverage is essential.

The selected model should balance the intended use's accuracy and operational requirements. A faster detector is not automatically better if it systematically misses a priority class. The present draft describes these interpretation criteria without asserting observed findings.

7.2 Comparison with literature baselines and existing solutions

The candidate architectures are grounded in the YOLO [1] and Faster R-CNN [2] research, but comparisons must use this study's samples and hardware. Published benchmark scores are contextual background rather than target achievements. The generic COCO dataset [4] likewise cannot substitute for water-specific test evidence.

The thesis should compare the trained models under the shared protocol and explain where the annotation policy limits comparability. A small colour-rule baseline for diffuse conditions may be revealed when a complex detector adds little value. Any supplementary experiment must be declared before final testing or clearly labelled exploratory.

7.3 Research contribution

The intended research contribution is a reproducible study of localized and diffuse contamination indicators in a common visual-screening workflow. The key planned artefacts are the class ontology, sample-wise dataset split, robustness analysis and severity evaluation. The method is an application and experimental contribution; no new detector architecture is currently proposed.

Actual contributions must be supported by released or examinable artefacts. A completed training run alone does not establish generalization, and a curated dataset should document both reference quality and collection bias.

7.4 Product contribution

The intended product contribution is a reusable image-analysis service that returns traceable boxes, labels and visual severity to the wider water-quality system. The overlay can help users inspect the

visual basis of a prediction. Its usefulness depends on valid coordinates, understandable qualified labels, stable interfaces and appropriate handling of low-quality inputs.

The component is designed to complement measurements and microscopy. It does not replace either analytical module or the final evidence-based interpretation.

7.5 Limitations and threats to validity

Limitations include ambiguous visual classes, imperfect annotations, limited independent samples, class imbalance, small target size, glare, changing backgrounds and differences between laboratory and field photographs. Container or background cues can become shortcuts if they correlate with class labels. Group-based splitting and background variation reduce this risk but cannot eliminate all domain shifts.

Box coverage is an imperfect severity proxy, and diffuse-conditioning grades may depend on subjective judgement. Photographs may not resolve tiny particles or establish material identity. A model trained on controlled samples should not be presented as validated for arbitrary rivers, wells or drinking-water supplies. Confidence calibration and out-of-distribution detection remain separate questions requiring evaluation.

7.6 Implications

If validated, the component could support supervised visual screening, documentation and prioritization of samples for further examination. Potential users include monitoring teams, laboratories and educational projects. Claims of improved public health outcomes or cost savings would require additional studies.

Sustaining the product requires model maintenance, reviewed image collection, hardware support and clear ownership of errors. Commercialization requires a separate assessment of user needs, dataset and model licences, service costs and applicable requirements. No market demand, business advantage or regulatory acceptance has been established.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis draft presents a proposed Contamination Detection component based on deep-learning computer vision. It defines how photographs will be annotated, how YOLO and Faster R-CNN candidates will be compared, and how boxes, class labels and qualified visual-severity outputs will be integrated into the group system.

The design explicitly separates discrete objects from diffuse appearances and distinguishes suspected plastic particles from confirmed microplastics. Its feasibility and performance remain to be demonstrated through the planned dataset, training and held-out evaluation. The document therefore concludes with a research design rather than a claim of completed experimental success.

8.2 Achievement of objectives

Table 8.1 records objective status. Specification in this draft is not equivalent to completion of the corresponding dataset, software or validation task.

Table 5: Objective status

Objective	Specified in draft	Remaining work
O1	Capture and annotation policy	Pilot collection and reviewer validation
O2	Dataset and splitting plan	Curate, annotate and freeze splits
O3	Model comparison protocol	Train, tune and evaluate both detectors
O4	Separate severity design	Collect ordinal labels and validate grading
O5	Interface and test plan	Implement integration and measure outcomes

8.3 Individual research and product contributions

The individual contribution currently documented is the component-specific framing, annotation design, architecture, severity method and evaluation plan. Planned deliverables are a reviewed dataset, reproducible training configuration, selected model, inference interface and complete evaluation record.

The final thesis should replace proposed ownership with evidence of the candidate’s actual work, including code revisions, annotation contributions and experiment logs, while acknowledging team and external assistance.

8.4 Future directions

Immediate work is to agree the capture setting and qualified class labels, collect a reviewed pilot dataset, test annotation consistency, and confirm sufficient visual resolution for each target. Implement the smallest complete inference workflow before expanding the dataset or tuning many models.

Future extensions may include segmentation for more accurate visible-area measurement, high-resolution tiling for small targets, a separately validated microscopic particle workflow, uncertainty calibration and field-domain adaptation. Each extension must be evaluated independently; adding a feature or higher resolution does not itself establish reliable microplastic detection.

REFERENCES

[1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/Redmon_You_Only_Look_CVPR_2016_paper.html. openaccess.thecvf.com

[2] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems*, vol. 28, 2015, pp. 91–99. [Online]. Available: https://proceedings.neurips.cc/paper/2015/hash/14bfa6bb14875e45bba028a21ed38046-Abstract.html. proceedings.neurips.cc

[3] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Computer Vision—ECCV 2014*, Lecture Notes in Computer Science, vol. 8693. Cham, Switzerland: Springer, 2014, pp. 740–755, doi: 10.1007/978-3-319-10602-1_48. Springer Nature Link

[4] World Health Organization, *Microplastics in Drinking-water*. Geneva, Switzerland: World Health Organization, 2019. Accessed: Oct. 11, 2026. [Online]. Available: https://www.who.int/publications/i/item/9789241516198.

[5] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010, doi: 10.1109/TKDE.2009.191. doi.org

[6] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125. [Online]. Available: CVF paper. openaccess.thecvf.com

[7] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2980–2988, doi: 10.1109/ICCV.2017.324. openaccess.thecvf.com

[8] P. F. Proença and P. Simões, "TACO: Trash annotations in context for litter detection," *arXiv preprint arXiv:2003.06975*, 2020. [Online]. Available: https://arxiv.org/abs/2003.06975. arxiv.org

[9] L. Biermann, D. Clewley, V. Martinez-Vicente, and K. Topouzelis, "Finding plastic patches in coastal waters using optical satellite data," *Scientific Reports*, vol. 10, Art. no. 5364, 2020, doi: 10.1038/s41598-020-62298-z.

GLOSSARY

Annotation: The assignment of reference labels and spatial boundaries to image objects or regions for model training and evaluation.

Average precision (AP): A measure summarizing the precision–recall relationship for a detection class under specified matching criteria.

Bounding box: A rectangle indicating the location and approximate extent of an object or visual condition within an image.

Class ontology: A documented set of target categories, their definitions and the rules governing their use in annotation.

Confidence score: A model-generated score indicating support for prediction. It does not directly represent contamination severity or water safety.

Convolutional neural network (CNN): A neural network that learns spatial features from images through convolutional operations.

Data augmentation: Controlled transformations applied to training images to increase variation while preserving valid labels and annotations.

Data leakage: The unintended sharing of information across training, validation and test partitions, potentially inflating evaluation results.

Diffuse visual condition: An appearance, such as cloudiness or discoloration, that extends across a region without clearly defined object boundaries.

Discrete object: An individually distinguishable target, such as a separable item of floating debris.

Domain shift: A difference between the data conditions encountered during training and those encountered during evaluation or deployment.

False negative: A reference target that the model fails to detect under the specified evaluation criteria.

False positive: A predicted detection that does not correctly match a reference target under the specified evaluation criteria.

Faster R-CNN: A two-stage object-detection framework that generates candidate regions before classifying objects and refining their bounding boxes.

Ground truth: Reviewed reference annotations used for training and evaluation, recognizing that annotations may contain uncertainty or error.

Hard negative: A non-target feature, such as glare, bubbles or container markings, that resembles a contamination indicator.

Held-out test set: An independent dataset reserved for final evaluation and excluded from training and model-selection decisions.

Inference: The application of a trained model to new input data to generate predictions.

Inference latency: The time required to produce predictions within explicitly defined processing boundaries.

Intersection over union (IoU): The intersection area of predicted and reference bounding boxes divided by their union area.

Localization: The identification of a target's position and spatial extent within an image.

Mean average precision (mAP): The average of class-specific average precision values, also averaged across IoU thresholds when specified by the evaluation protocol.

Non-maximum suppression (NMS): A post-processing method that reduces redundant overlapping detections according to confidence scores and an overlap threshold.

Object detection: A computer-vision task that identifies target categories and locates their positions within an image.

Precision: The proportion of predicted detections that correctly match reference targets.

Recall: The proportion of reference targets correctly detected by the model.

Region of interest (ROI): The selected area of an image containing the water sample to be analyzed.

Sample-wise splitting: Partitioning a dataset while keeping all images of the same physical water sample within a single training, validation or test partition.

Suspected plastic particle: A sufficiently resolved particle whose appearance suggests plastic but whose material composition has not been independently confirmed.

Transfer learning: The adaptation of knowledge or model parameters learned from a source dataset to a target task.

Validation set: A dataset used to select model configurations, checkpoints and detection thresholds before final testing.

Visual contamination indicator: An observable image feature associated with possible contamination, interpreted within the limitations of photographic evidence.

Visual severity: A defined grade describing the visible extent or appearance of a contamination indicator, distinct from prediction confidence, contaminant concentration and health risk.

YOLO (You Only Look Once): A family of single-stage object detectors that predict object locations and categories through a unified detection network.

APPENDICES

Appendix A: Research Instruments and Proposed Test Cases

The following instruments support the proposed data collection, annotation and evaluation procedures. They do not represent completed experiments or verified results.

Table 6: Image Acquisition and Annotation Record

Field	**Information to record**
Sample ID	Unique identifier for the physical water sample
Image ID	Unique identifier for the photograph
Capture conditions	Device, lighting, background, container, viewing distance and angle
Image properties	Original dimensions, file format and image-quality observations
Data provenance	Image source, permission and applicable licence
Region of interest	Coordinates defining the visible water region
Contamination label	Category assigned according to the annotation guide
Bounding box	Target coordinates in the original image
Annotation uncertainty	Ambiguous appearance, occlusion or insufficient resolution
Reviewer details	Reviewer identifier, annotation date and guide version

A.2 Annotation and Review Procedure

A pilot subset will be annotated independently by two reviewers. Disagreements concerning target categories, bounding-box boundaries and uncertain appearances will be documented and resolved before the annotation guide is finalized. Discrete objects, such as floating debris, will be distinguished from diffuse conditions, such as cloudiness and discoloration. Suspected plastic particles will retain qualified labels unless their material identity is independently confirmed.

Reference visual-severity grades will be assigned using a class-specific rubric established during the pilot stage. Grading will remain independent of model confidence and predicted bounding boxes. Images that cannot support reliable grading will receive an “unavailable” status with a recorded reason.

Table 7: Proposed Test Cases

Test ID	Test condition	Expected behaviour or assessment
A01	Valid image submitted	Return a structured response containing the correct sample and image identifiers.
A02	Corrupt or unsupported image	Reject the input with an explicit error status.
A03	Image exceeds configured size limits	Reject the input before resource-intensive processing.
A04	Resizing or padding applied	Restore predicted boxes to original-image coordinates within a defined tolerance.
A05	Overlapping boxes used for coverage	Calculate union coverage without double-counting overlapping regions.
A06	Diffuse condition occupies the full water region	Apply the separate severity method without automatically assigning high severity.
A07	No detections retained	Return a no-detection status without implying that the water is safe.
A08	Inference failure or timeout	Return an explicit failure status linked to the original sample.
A09	Findings transferred to the reporting component	Preserve qualified labels, confidence scores, severity and model version.
A10	Glare, bubbles or container markings present	Measure false-positive detections using labelled challenging negative examples.

A.4 Experimental Results Record

For each experiment, record the dataset version, sample-wise split, model architecture, pretrained checkpoint, training configuration, random seed and hardware. Evaluation records will include per-class precision, recall, average precision, false-positive rates, severity agreement and inference latency. Results for discrete objects and diffuse conditions will be reported separately. Numerical results and test outcomes will be added only after the relevant experiments have been completed.

Appendix B: Academic Integrity and AI Use Disclosure

Generative AI was used to assist with drafng, restructuring and language refinement of thesis content, including the abstract, literature-review passages, glossary and appendix materials. It also assisted in identifying candidate references and formatting citations.

The candidate is responsible for verifying the accuracy of AI-assisted content, checking references against original publications and ensuring that the thesis reflects the research conducted. No generated numerical values are presented as experimental results. The final disclosure will document the tools and assistance used in accordance with institutional requirements.